

Physics Research Paper

Effect of Dust accumulation on the power output of a photovoltaic solar cell.

Research Question: How does dust accumulation on the surface of a solar cell affect its electrical output ?

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1. Introduction

I have always been into machinery and objects that use electrical energy, and I used to play with toys like remote controlled cars, motors, and several other toys combined with electrical components. What primarily struck my interest in the system of a solar cell was the way it operates, by inputting solar energy from the photons, and converting it into electrical energy to supply electricity to huge grids and homes (**Alternative Energy Tutorials**). As I began to explore a broader aspect of renewable energy and power-system engineering, photovoltaic systems seemed crucial and interesting to me, primarily due to their increasing popularity around the world as countries transition toward sustainable energy infrastructure. A recent discussion that I had with my dad regarding the use of photovoltaic cells (PV) farms in high-illuminance regions, especially in semi-arid desert environments, where huge land availability makes it ideal for solar installations.

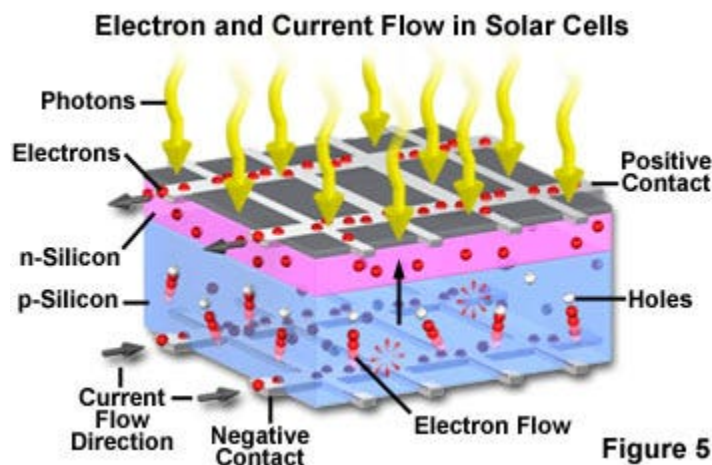
Additionally, we emphasized that a critical challenge had been increasingly observed across global PV implementations, dust accumulation. This problem primarily strikes on the surface of the solar modules and it significantly reduces the transmittance of incoming solar radiation, altering the optical properties of the protective glass layer and consequently decreasing the electrical output of PV cells (Rashid et al.). It reduces transmittance primarily by scattering and absorbing photons, reducing the number of electron-hole pairs generated inside the semiconductor. In certain regions with minimal rainfall and high particulate concentration, dust irregularity has become one of the most important factors limiting long term PV continuity, often causing frequent cleaning cycles and reducing the energy yield. This emerging issue struck my motivation behind the investigation on how varying dust quantity on a solar cell surface would influence the voltage, current and overall power output determined by the cell, forming the basis of my research question that guides this investigation.

1.1 Theoretical background

The photovoltaic effect is a fundamental principle that enables the solar cells to convert the solar energy into electrical energy. This fascinating conversion takes place when incident photons (light particles) strike the semiconductor junction, causing the excitement of electrons into the conduction (collection plate) band, creating electron-hole pairs that produce a vast amount of electromotive force and hence deliver current to an external circuit (**Alternative Energy Tutorials**). In real life systems, the environmental factors contribute a lot of challenges that modifies the irradiance reaching the cell surface. Because the dust particles can essentially:

- Scatter incoming photons
- Absorb part of the incident radiation
- Reflect light away from the cell
- Physically block incident light from reaching the semiconductor surface.

This reduces the effective irradiance (E) reaching the photovoltaic material. Since the cell's output is heavily concentrated on the received light intensity, any reduction in the incident irradiance due to the dust deposition will decrease the generated current and overall electrical power. Thus, being able to understand the relationship is vital in order to improve the photovoltaic module design especially in regional placements where the atmospheric particulate concentration is high.



(Figure 1: Solar cell photon-electron Reference Image; Bibliography #8)

Traditional formulas used to determine electrical power output and input:

- $P_{out} = I \times V$
- $P_{in} = E \times A$
- $I = \frac{P}{V}; V = \frac{P}{I}$
- $P = \frac{V^2}{R}$

E : Irradiance (W/m^2)

A: Area of the PV cell.

Irradiance and current are directly proportional to each other, meaning that current is linearly dependent on irradiance for a fixed load. The current generated by the photovoltaic cell is directly proportional to the irradiance incident on its surface. Meaning that if the light intensity increases, larger amounts of photons strike the semiconductor material, producing more electron-hole pairs. Additionally, increased temperature will decrease the bandgap of silicon, lowering the open-circuit voltage.

1.2 Research question

“How does dust accumulation on the surface of a solar cell affect its electrical output ?”

1.3 Rationale

Since modern photovoltaic panels utilize protective glass covers for weather resistance, protection and durability, these surfaces are not 100% immune to dust deposition. Due to the rise in challenges for the solar cell industry, perpetuated by dust accumulation on the surfaces, this investigation solely aims to quantify how dust mass essentially affects core performance metrics of a solar cell, providing insight relevant to large-scale renewable energy implementations. Hence this ignites my investigation as it bridges my interest in renewable energy engineering, with a practical challenge faced by real world applications.

1.4 Hypothesis

Increasing the dust mass and spread on the surface of the photovoltaic cell, will reduce the electrical output. Higher dust mass and effective coverage will prevent more incident photons, lowering the generated current, thereby reducing the overall electrical power produced by the cell. (Rashid et al.)

2. Materials, Variables & Methodology

2.1 Variable declaration

Variable Type	Explanation	Apparatus (Materials)
Independent Variable	Talcum powder is applied on the solar panel's surface, will be used with 3 trials each and due to that the power output varies in each trial. I will be putting a particular weighted amount of talcum powder.	Talcum Powder - 0 - 5g - 10g - 15g - 25g
Dependent Variable	- Voltage (V) - Current (A) - Power output (W)	For voltage and current, a multimeter is used.

Table 1: Variable Types and variables

Controlled Variables		
Variables	Significance	Means of control
Illuminance (lux)	Monitored using a lux meter to ensure constant incident light intensity.	Controlled by using the same lamp throughout the entire experiment.
Angle of the solar cell	The angle of the solar cell must be kept constant so it receives an equal amount of irradiance at each trial.	Keep 180° flat below the lamp source.
Solar cell used	Using different solar panels might result in redundant power and voltage outputs.	Ensuring that a different solar cell is not used.

Controlled Variables		
	Thereby, to mitigate this irregularity, the solar cell must be consistent throughout the entire investigation.	
Lamp distance	Distance between the solar panel and the lamp must be kept constant to maintain an equal amount of light intensity distribution, hence to maintain constant illuminance reaching at the panel's surface.	Maintain 15cm
Dust Type (talcum powder)	Maintaining the same dust type prevents large deviations in achieved data from the solar panel's output reports.	Use one single talcum powder type with different masses.

Table 2: Controlled Variables

Uncontrolled Variables	
Variables	Explanation
Environmental restrictions.	The investigation is based inside a packed room, however there may be temperature fluctuations causing higher electron flow, eventually decreasing voltage. It should have a negligible impact but there would be minor discrepancies due to the temperature differences.

Table 3: Uncontrolled Variables

2.2 Methodology

1. Use a scientific weighing scale to determine the mass of each powder sachet.

2. Place the solar panel at 180° directly below the lamp at the point of maximum irradiance.
3. Turn on the lamp and warm it up for 3 minutes before starting.
4. Use the lux meter to determine and ensure that the illuminance is constant throughout the experiment.
5. Measure the surface area of the solar panel that contributes towards calculating the power input.
6. **Prepare** all masses of talcum powder from 0g to 25g.
7. **For 0g**, measure and record the voltage and current, doing this **3 times**.
8. **For 5g**, spread the powder equally across the panel. Measure and record the voltage and current. Clean the panel with a soft cloth, then repeat this process for a **total of 3 trials**.
9. **For 10g**, spread the powder equally across the panel. Measure and record the voltage and current. Clean the panel with a soft cloth, then repeat this process for a **total of 3 trials**.
10. **For 15g**, spread the powder equally across the panel. Measure and record the voltage and current. Clean the panel with a soft cloth, then repeat this process for a **total of 3 trials**.
11. **For 20g**, spread the powder equally across the panel. Measure and record the voltage and current. Clean the panel with a soft cloth, then repeat this process for a **total of 3 trials**.
12. **For 25g**, spread the powder equally across the panel. Measure and record the voltage and current. Clean the panel with a soft cloth, then repeat this process for a **total of 3 trials**.
13. Calculate **power output** manually for each dust mass, using the voltage and current values determined from the multimeter.

2.2.1 Structure diagram

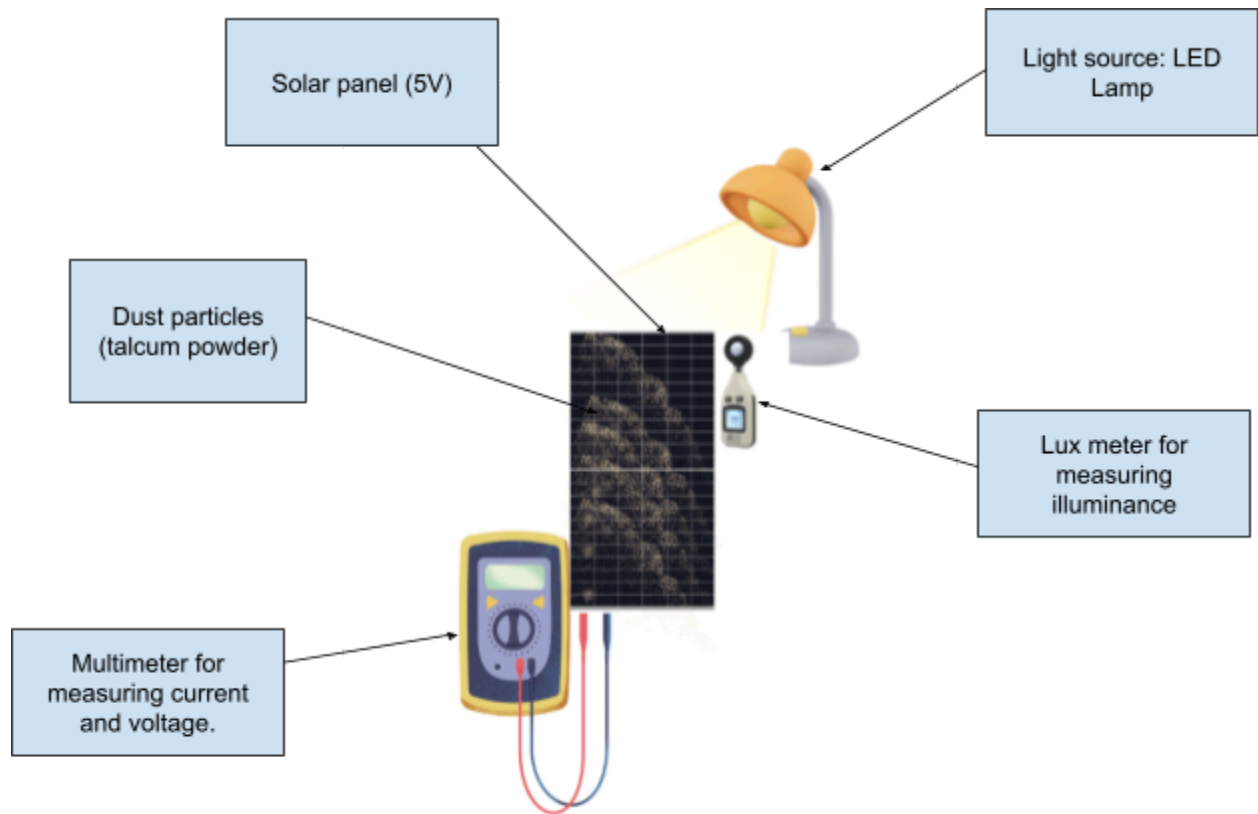


Figure 1: Digital diagram of the experiment

This structure diagram represents how the actual investigation will take place with varying masses of talcum powder.

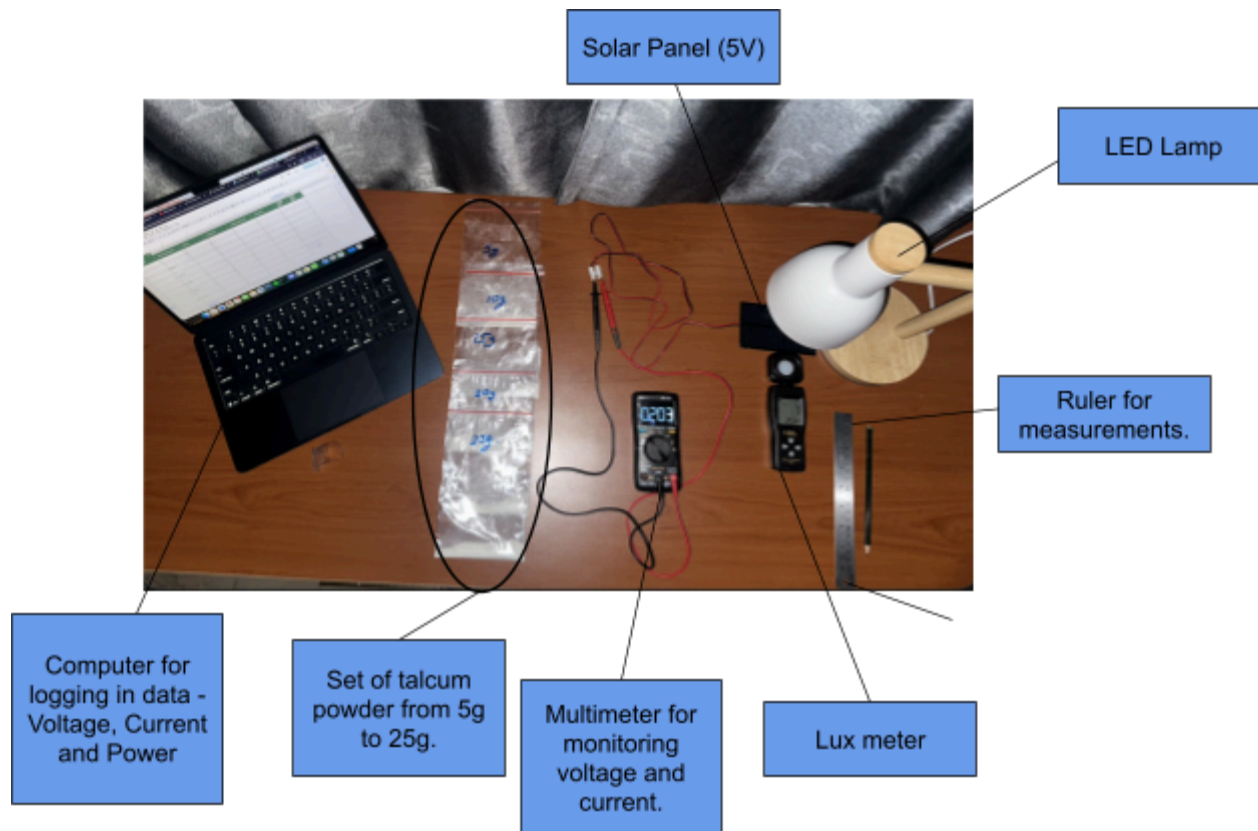


Figure 2: Experiment setup

2.2.3 Equipments

- Solar Panel x 1
- Lux meter x 1
- Talcum powder x 0g, 5g, 10g, 15g, 20g, and 25g.
- LED Lamp x 1
- Multimeter x 1

2.3 Risk Assessment

Consideration Type	Importance
Ethical Considerations	<i>Not Applicable</i>
Safety Considerations.	This investigation comprises minimal exposure to sunlight avoiding physical injury. The lamp operates at low voltage and does not produce intense heat. All the electrical items were insulated every trial.
Environmental Considerations	All procedures were conducted indoors under controlled conditions. The talcum powder is safe for handling and disposal. No materials were released into the environment, and minimal energy was consumed by the LED lamp.

Table 4: Risk Assessment table

3. Data collection

3.1 Actual data extraction - Lamp source

The illuminance (L) was measured using the lux meter. The irradiance (E) was then estimated by the assumption of a constant luminous efficacy (k) for the white LED lamp that was used.

- $E = \frac{L}{k}$
- $E = \frac{2608}{100} \approx 26.08 \text{ W m}^{-2}$

L: Illuminance

k: Luminous Efficiency constant

Measurements of the Solar Panel :

- $11 \text{ cm} \times 7 \text{ cm}$
- $\text{Area} = 77 \text{ cm}^2 \Rightarrow 0.0077 \text{ m}^2$

Raw Data Collection from Excel

Dust mass (g)	Trial (s)	Voltage (V) $\pm 0.05V$	Current (mA)	Current (A) ± 0.0000 3 A	Power output (W)
0	1	3.8	1.48	0.001480	0.005624
	2	3.8	1.48	0.001480	0.005624
	3	3.8	1.47	0.001470	0.005586
5	1	3.31	0.61	0.000770	0.0025487
	2	3.28	0.6	0.000600	0.001968
	3	3.24	0.64	0.000640	0.0020736
10	1	2.897	0.33	0.000330	0.00095601
	2	2.861	0.33	0.000330	0.00094413
	3	2.739	0.3	0.000300	0.0008217
15	1	2.308	0.12	0.000120	0.00027696
	2	2.219	0.12	0.000120	0.00026628
	3	2.196	0.1	0.000100	0.0002196
20	1	1.544	0.04	0.000040	0.00006176
	2	1.542	0.04	0.000040	0.00006168
	3	1.52	0.04	0.000040	0.0000608
25	1	1.249	0.02	0.000020	0.00002498
	2	1.202	0.02	0.000020	0.00002404
	3	1.19	0.02	0.000020	0.0000238

Table 5: Raw Data

Data Processing:

Worked example for calculating **Power**, with 0 grams of powder.:

- $P = I \times V$
- $P = (0.001480)A \times 3.8V$
- $P = 0.005624W$

Fractional uncertainty:

- $\frac{\Delta P}{P} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$
- $\frac{\Delta V}{V} = \frac{0.05}{3.8} = 0.01316$
- $\frac{\Delta I}{I} = \frac{0.00003}{0.001480} = 0.02027$

$$\therefore \frac{\Delta P}{P} = 0.01316 + 0.02027 = 0.03343$$

Absolute uncertainty for worked example (Power):

- $P_{out} = P(0.03343)$
- $\therefore (0.005624)(0.03343) = 1.88 \times 10^{-4}W$
- $\therefore P_{out} = (5.624 \pm 0.188) \times 10^{-3}W$

3.2 Data analysis

Mean calculation formula for each quantity (V, I, and P):

- $\bar{V}_{per\ mass} = \frac{v_1 + v_2 + v_3 + v_4 + \dots + v_n}{n}$
- $\bar{I}_{per\ mass} = \frac{i_1 + i_2 + i_3 + i_4 + \dots + i_n}{n}$

- $\bar{P}_{per\ mass} = \frac{P_1 + P_2 + P_3 + P_4 + \dots + P_n}{n}$

The calculations were established through the TI Nspire GDC, for each mean value, along with determining Power (P)'s uncertainty.

Additionally, to calculate the uncertainty for Mean power values, I must use the varying values of $\bar{V}$ and $\bar{I}$:

$$\frac{u_{\bar{P}}}{\bar{P}} = \frac{u_{\bar{V}}}{\bar{V}} + \frac{u_{\bar{I}}}{\bar{I}}$$

$$u_{\bar{P}} = \bar{P} \left(\frac{u_{\bar{V}}}{\bar{V}} + \frac{u_{\bar{I}}}{\bar{I}} \right)$$

$u_p \Rightarrow$ **Absolute Uncertainty** for $\bar{P}$

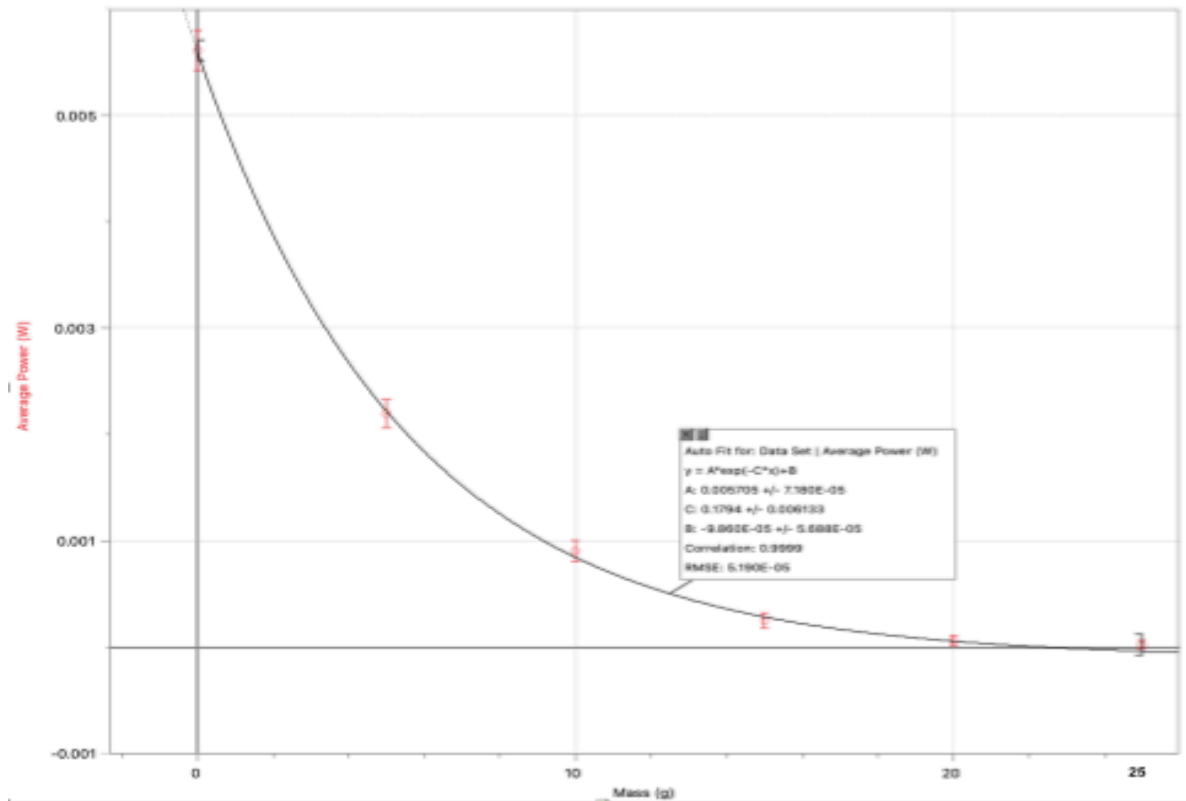
Sample Calculation of Mean Power ($\bar{P}$) Uncertainty

$$u_{\bar{P}} = 0.005611 \left(\frac{0.05}{3.8} + \frac{0.00003}{0.00148} \right) = 0.000188$$

Mass (g)	Mean Voltage ($\bar{V}$) $\pm 0.05V$	Mean Current ($\bar{I}$) $\pm 0.00003A$	Mean Power ($\bar{P}$) W
0	3.8	0.00148	0.005611 ± 0.000188
5	3.276	0.00067	0.002197 ± 0.000132
10	2.832	0.00032	0.000907 ± 0.000101
15	2.241	0.000113	0.000254 ± 0.000073
20	1.535	0.00004	0.000061 ± 0.000048
25	1.213	0.00002	0.000024 ± 0.000037

Table 6: Mean data collection

For the following graph I used **Logger Pro**, by inputting all sets of mean Power determined in the table 6 (y - axis) and the increasing mass of the talcum powder (x - axis):



Graph 1.1 Average Power. vs Mass.

Based on the graph obtained from logger pro, we can see that the relationship between the mean power and increase in mass, is denoted by the **exponential function**. Where the power exponentially decreases as the mass increases.

The graph represents the following exponential function:

$$\bar{P} = Ae^{-Cm} + B$$

B is an extremely small value which can be neglected for the sake of this graph.

Finalized exponential function:

$$\bar{P} \approx Ae^{-Cm}$$

$$\bar{P} \approx (0.005705)e^{-0.1794m}$$

The constant **A** represents the theoretical power output when no dust is present, with the mass as 0. Comparing this to the actual measured value of power at 0g dust: $P_0 \approx 0.005611$ and the theoretical value of **A** being 0.005705, there is a slight deviation of about 1.7%. This closeness of the result validates the exponential model's accuracy, as the fitted curve accurately plots and predicts the clean panel performance. This minor deviation falls well within the experimental uncertainty range, representing that the model parameter **A** reliably represents the baseline power output. The experimental value lies within the uncertainty bounds of the theoretical prediction, indicating no systematic error in the modeling approach.

Furthermore, the **decay constant C, 0.1794**, further quantifies the sensitivity of the solar cell to dust concentration. It essentially represents the rate at which power output decreases per gram of added dust.

Focusing on the quantitative power loss, the endpoints of the investigation reveal the intensity of **dust accumulation effects**. Initially with a clean panel the power output determined was 5.611 mW whereas the final one after using 25g of powder, was 0.024mW which retained a loss of 5.587 mW. Considering a percentage reduction, that would be approximately 99.6% loss in the power output with only 25g of powder. In essence the extreme amount of power loss demonstrates the extreme sensitivity of photovoltaic cells to surface contamination. Even the lowest dust mass tested resulted in approximately 60.8% power loss. This indicates that dust impact is most severe during the energy conversion process with diminishing marginal effects as coverage approaches saturation.

Now let us **linearize** the exponential function, so that we can develop a linear modelled graph. Considering a linearized graph, it enables me to implement **maximum** and **minimum error** lines. In order to linearize, I must convert the exponential function into a logarithmic function:

$$\ln(\bar{P}) \approx \ln(A) - Cm$$

C: Attenuation Constant

The following table consists the logarithmic mean power and mass values **only**:

Mass (g/x-axis)	Mean power ($\bar{P}$) / (W/y-axis)	$\Delta(\ln(P))$
0	$\ln(0.0056110)$	0.0335
5	$\ln(0.002197)$	0.0601

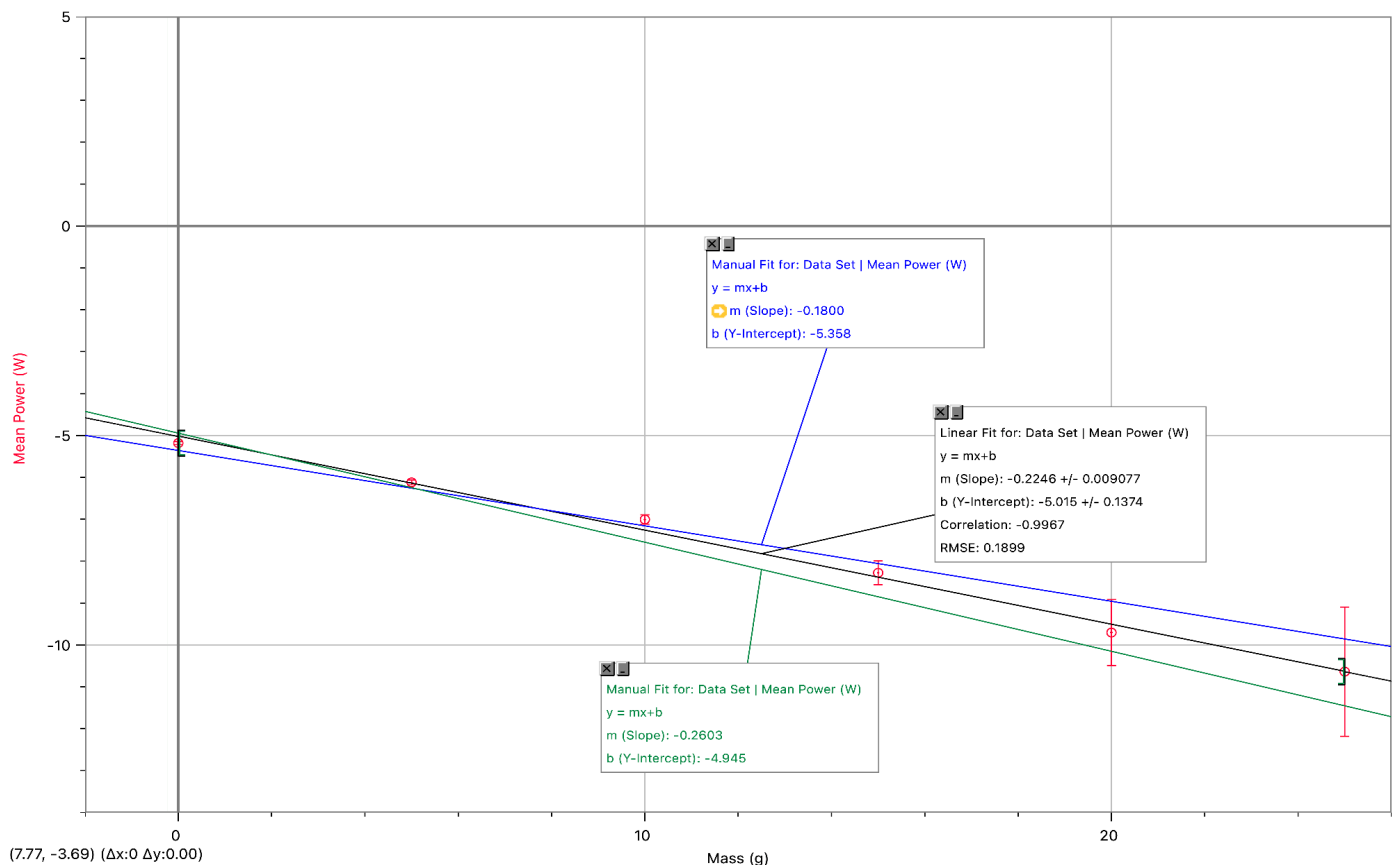
Mass (g/x-axis)	Mean power ($\bar{P}$) / (W/y-axis)	$\Delta(\ln(P))$
10	$\ln(0.000907)$	0.1114
15	$\ln(0.000254)$	0.2874
20	$\ln(0.000061)$	0.7869
25	$\ln(0.000024)$	1.542

Table 7: Linearized set of values

Error Propagation:

This error method is used to derive the new set of uncertainties considering logarithmic values, as it contributes to the error bars on the graph.

- $y = \ln(P)$
- $\Delta y = \left| \frac{d(\ln(P))}{dP} \right| \times \Delta P$
- $\Delta(\ln(P)) = \frac{\Delta P}{P}$



Graph 1.2 Linearized Function

Uncertainty calculation in the slopes:

$$\Delta m = \frac{|m_{max} - m_{min}|}{2}$$

$$\Delta m = \frac{|-0.1800 - (-2.603)|}{2}$$

$$\Delta m = \frac{0.0803}{2}$$

$$\Delta m = 0.04015$$

Therefore, the **gradient** of the new linearized graph is:

$$\Delta m = 0.04015 \pm 0.0402$$

The new linearized graph **still** shows a **declining** relationship between the variation of mass concentration and power output of the solar cell. Comparing it with the exponential decay function determined earlier, there is a similarity between the two, that both represent a declining state of the power output as the dust mass increases. Contrasting the slopes or the decay constant of both, exponential and linear, vary:

$$m_{linear} = -0.2246 \pm 0.0402$$

$$C = -0.1794$$

The difference between the attenuation **constants** achieved from the exponential regression and the linearised model, arises primarily from how uncertainty varies under the logarithmic transformation. Since ΔP remains roughly similar in magnitude, the ratio $\frac{\Delta P}{P}$ becomes large. This essentially causes the error bars in the linearised graph to increase significantly at higher mass values.

In contrast, the nonlinear exponential regression directly fits the actual, and original exponential model to the raw data. It doesn't **distort uncertainties** due to logarithmic deviation. Instead, it uses statistical fitting to determine the best value of C by minimising the residual error across all data points. Therefore, although both methods confirm decay in the relationship, the nonlinear regression provides a more accurate and robust estimate of the attenuation constant.

4 Conclusion

In conclusion this investigation examined how increasing dust accumulation affects the electrical output of a photovoltaic solar cell. The results show a clear negative relationship, as dust mass increased, both current and voltage decreased, leading to a strong reduction in overall power output. The data followed an exponential decay trend rather than a linear one, indicating high sensitivity of the solar cell to surface contamination.

This behavior is physically consistent with photovoltaic theory as stated in the introduction. Dust particles scatter, absorb, and block incident photons, which causes heavy reduction in the effective irradiance reaching the semiconductor. Since current is directly proportional to irradiance, reduced light intensity directly lowers charge carrier generation and therefore electrical power output.

Both nonlinear regression and linearized analysis confirmed the declining trend, despite minor differences in calculated decay constants due to uncertainty amplification in the logarithmic transformation. In essence the obtained results from the experiment support the hypothesis and demonstrate that dust accumulation significantly reduces photovoltaic performance, highlighting the importance of surface maintenance in real-world solar installations.

5 Evaluation

Although the exponential model closely fits the experimental data with a correlation of 0.9999, several limitations influence the calculated attenuation constant $C = 0.1794/\text{g}$. As the dust mass increases and current approaches the multimeter's resolution limit, the fixed uncertainty of $\pm 0.00003\text{A}$ causes fractional uncertainty in power to grow from 3.3% at 0g, to over 150% at 25g. This distorts the linearized model, explaining the divergence between the slope of -0.2246 and the nonlinear decay constant of -0.1794. The nonlinear regression remains statistically stronger as it fits and aligns directly to raw data without logarithmic manipulation.

5.1 Weaknesses and Limitations

Considering this information, additional **human** and **systematic** errors played a significant role in the **errors** and **uncertainties calculated**.

Source	Impact	Improvement
Human errors		
Non uniform dust distribution	Irregular distribution of powder for each mass as a card was used for it. This lacks precision	Controlled Distribution: Use a fine mesh sieve to evenly distribute across the solar panel.
Intermediate Mass Values	Large intervals between the dust mass values resulted in fewer data points, producing a lower-resolution trend in the graph.	Use smaller incremental dust mass values to obtain more data points and produce a more accurate representation of the relationship between dust accumulation and power output.

Table 8: Weakness table for Human errors

Source	Impact	Improvement
Systemic errors		
Assumption of Constant Irradiance	I assumed from the beginning of the investigation that the irradiance remains consistent however LED output can definitely fluctuate with temperature.	Controlled Irradiance: Use a lux meter to monitor the light intensity and allow the LED to warm up before trials to maintain stable irradiance.

Table 9: Weakness table for Systemic errors

5.2 Strengths

Strength	Discussion
Standardized number of trials	This investigation used 3 trials for each mass value, enabling variance between trials to be easily observed while improving the reliability of collected data.
Preliminary Trial	Taking a trial before implementing any powder, allowed me to determine the original and authentic output of the solar panel, which enables a distinct comparison with added powder.
Controlled Variables	Variables such as the positioning of the panel, same solar panel, and identical dust type were kept constant to ensure the results only reflected the effect of dust mass, rather than other inconsistencies.

Table 10: Strengths Table

5.3 Extension

Hence based on the determined results, a possible solution to this global problem is to implement automatic cleaning systems with sensors that detect dust accumulation levels and trigger cleaning without human intervention. This would help maintain maximum light exposure on the solar panel surface and sustain higher electrical power output, making the solar panels much more efficient.

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